



PRV NURSE

A Home Nursing Access and Placement Platform

Connecting Qualified Unposted Nurses in Ghana to Home-Care Jobs

A Thesis Presented to the University of Energy and Natural Resources in
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CHAPTER 1: INTRODUCTION

Abstract

Ghana faces a widening gap between the supply of trained nurses and the demand for home-based medical care. Many qualified nurses complete their training but remain unposted, while families caring for elderly relatives, post-surgical patients, and individuals with chronic conditions struggle to find vetted, professional caregivers at home. Existing informal arrangements lack standardization, credential verification, and reliable scheduling, which creates safety risks for patients and unpredictable income for nurses.

This thesis proposes PRV Nurse, a Home Nursing Access and Placement Platform designed to bridge this gap. The platform enables verified nurses to create professional profiles, set availability and pricing, and receive bookings from patients or caregivers. A machine learning recommendation engine matches patients to the most suitable nurses based on skills, location, availability, and historical performance data. The system is built on a React and Tailwind CSS frontend, a Node.js REST API backend, and a MongoDB database. An administrative dashboard provides oversight of nurse verifications, bookings, payments, and platform health.

The proposed architecture follows a modular, role-based design covering three primary user types: nurses, patients, and administrators. Key system components include a nurse onboarding and verification pipeline, a patient care request module, a smart matching engine, a scheduling and booking system, a mobile money payment integration, and a care documentation module. This paper covers the introduction, literature review, and methodology, with architecture diagrams, data flow models, and comparative analysis to demonstrate the design decisions.

Keywords: Home nursing, placement platform, credential verification, machine learning, nurse-patient matching, Ghana, Node.js, React, mobile health.

1.1 Background

The delivery of health care has gradually expanded beyond the walls of hospitals and clinics into patients' homes. Advances in medical technology, the increasing burden of chronic diseases, and the preferences of aging populations have made home-based care not just convenient but, in many cases, medically preferable [1]. Patients recovering from surgery, managing diabetes, or living with mobility limitations benefit greatly from professional nursing support delivered at home, reducing both hospital readmission rates and the cost of care [2].

In Ghana, this transition toward community and home-based care is still in its early stages. The country's formal health system remains heavily facility-oriented, with limited infrastructure for coordinated home nursing. At the same time, demographic trends are shifting. The Ghana Statistical Service reported in its 2021 Population and Housing Census that Ghana's elderly population is growing steadily, and chronic disease burdens, particularly hypertension, diabetes, and cardiovascular conditions, are on the rise [3]. These trends point to a growing and unmet demand for home nursing services.

On the supply side, Ghana produces a significant number of trained nurses annually through its nursing colleges and universities. However, due to budgetary constraints and hiring freezes in the public health sector, many graduates complete their training and licensing without receiving government postings. These nurses are professionally qualified but economically inactive, representing a significant waste of trained human capital [4]. They are willing and able to work but lack the channels to connect with patients who need their skills outside of formal facility settings.

Digital platforms have proven in other contexts that they can efficiently match supply and demand in service markets. Ride-hailing applications matched idle drivers to passengers. Freelancing platforms matched skilled workers to short-term clients. Similar logic applies to nursing: a well-designed digital marketplace can match unposted qualified nurses to families and patients seeking reliable home care, provided that trust, verification, and accountability mechanisms are built in [5].

PRV Nurse is a proposed home nursing access and placement platform that addresses this gap. It is designed specifically for the Ghanaian context, incorporating mobile money payments, GPS-based matching, institutional referral-based credential verification, and a machine learning recommendation system that improves match quality over time.

1.2 Problem Statement

Three persistent gaps limit access to reliable home nursing in Ghana and motivate this work:

- **Disconnected matching:** qualified nurses and patients who need them exist in the same communities but have no efficient way to find each other. Word-of-mouth referrals and informal arrangements are the primary discovery channels, which are slow, limited in reach, and unreliable. This gap results in unposted nurses staying idle and patients receiving substandard or informal care [4].
- **Unclear quality and safety:** without standardized credential checks, background screening, or service documentation, clients have no reliable way to assess whether a caregiver is properly trained. Conversely, nurses have no standardized way to demonstrate their qualifications and build a professional reputation outside of public institutions [6].
- **Operational friction:** scheduling, payment, and documentation in informal home-care arrangements are mostly verbal or paper-based, which leads to disputes, poor follow-through, and no continuity of care records for patients. This lack of structure discourages professional nurses from operating in the home-care market [2].

Taken together, these gaps mean that patients who would benefit from professional home care either go without it or settle for unqualified helpers, while trained nurses remain unemployed. A structured digital platform that addresses all three gaps simultaneously can create real social and economic value.

1.3 Objectives

The main objective of this project is to design and implement a home nursing placement platform that creates ethical, sustainable employment for qualified nurses in Ghana while improving access to safe, professional home-care services. The specific objectives are:

1. To design a nurse onboarding and verification workflow that validates credentials, license status, and institutional referrals before a nurse is activated on the platform.
2. To build a patient request and care requirement module that captures detailed care needs, medical history, preferred schedule, and location for accurate matching.
3. To implement a machine learning recommendation engine that matches patients to suitable nurses based on skills, proximity, availability, ratings, and personalized preference signals.
4. To develop a scheduling, booking, and mobile money payment system that manages the full service lifecycle from request to payment.
5. To create an administrative dashboard for overseeing nurse verifications, managing bookings, monitoring platform health metrics, and resolving disputes.
6. To evaluate the proposed system using prototype testing, simulated matching scenarios, and a planned pilot study.

1.4 Scope and Delimitations

This project is scoped to home nursing services in urban and peri-urban Ghana, with an initial focus on the Greater Accra Region. The services covered are those within the professional scope of registered nurses, including medication administration, wound care, vital signs monitoring, elderly care support, post-surgical care assistance, and chronic disease monitoring. The platform does not replace hospital care and does not cover services requiring surgical or specialist medical intervention.

The system assumes that users have basic mobile phone access and data connectivity. Payment processing relies on existing mobile money infrastructure, specifically MoMo-based services. The platform does not handle payments directly but provides a structured interface and transaction records through integration with mobile money APIs.

Emergency cases that require ambulance or hospital admission are explicitly outside the scope of platform-mediated care and are escalated to appropriate emergency services through clearly defined protocols.

1.5 Significance of the Study

This work contributes to several important areas. Economically, it creates a formalized earning channel for qualified nurses who are currently unproductive, converting trained human capital into active, income-generating professionals. Socially, it expands access to professional nursing for patients who cannot easily travel to facilities, particularly the elderly and those recovering from surgery [1].

From an engineering and systems design perspective, the project demonstrates how a multi-sided platform with trust mechanisms, machine learning matching, and mobile-native design can be built for a low- and middle-income country (LMIC) context. The architecture decisions, trade-offs, and evaluation methodology developed in this project contribute to the growing body of knowledge on digital health platforms in sub-Saharan Africa [7].

Finally, the use of machine learning for nurse-patient recommendation in a resource-limited context is a technically novel contribution. Most existing platforms in this space rely on simple rule-based filtering; this project investigates how collaborative and content-based filtering techniques can improve match quality and personalization over time [8].

1.6 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 provides a literature review covering home care platforms, digital health marketplaces, credential verification systems, machine learning in healthcare recommendation, and mobile health in sub-Saharan Africa. Chapter 3 presents the system methodology, including the overall architecture, data models, matching algorithm design, technology stack decisions, and evaluation plan. The References section lists all cited works in IEEE format.

CHAPTER 2: LITERATURE REVIEW

2.1 Home-Based Care: Context and Trends

Home-based care refers to the delivery of medical or supportive services within a patient's residence rather than at a health facility. The World Health Organization (WHO) defines home-based long-term care as services that support people to live at home and in their communities while maintaining health and well-being [1]. Globally, home care is growing as a preferred model due to patient preference, cost reduction, and improved outcomes for specific conditions such as chronic illness management and post-surgical recovery [9].

In sub-Saharan Africa, the formal home care sector is underdeveloped. Most care is delivered informally by family members, untrained community workers, or informally engaged helpers. Ssengooba et al. [10] found that the lack of trained home-care workers in East Africa was a major barrier to scaling care for elderly and chronically ill populations. Similar patterns have been documented in Ghana by Agyemang [11], who noted that families in Accra relied heavily on informal networks rather than professional services, not because professional nurses were unavailable but because there was no accessible mechanism to connect to them.

This mismatch between trained supply and unmet demand is the fundamental problem PRV Nurse addresses. Literature on service marketplaces consistently shows that when proper information channels, trust mechanisms, and transaction support are provided, both supply and demand activate at higher rates [12]. The key is designing those mechanisms to fit the local context.

2.2 Digital Health Marketplaces and Platforms

Digital health platforms have emerged as a category of technology that connects health service providers to consumers, often bypassing traditional institutional intermediaries. Klasnja and Pratt [13] provide a foundational taxonomy of digital health applications, identifying care coordination platforms as a major category with demonstrated impact on access and continuity of care.

Several platforms worldwide have demonstrated the viability of nurse and caregiver marketplaces. Care.com in the United States and Homage in Southeast Asia are examples of platforms that match caregivers to clients, handle scheduling, and process payments [14]. However, these platforms operate in markets with established banking infrastructure, well-functioning credential databases, and strong consumer protection regulations. Adapting their model to Ghana requires significant re-engineering of the trust and payment components.

Research by Meingast et al. [15] on healthcare platform design emphasizes that the core challenge in any health marketplace is not the matching algorithm but the trust infrastructure. Users, both patients and nurses, need confidence that the platform has verified credentials, handles personal and medical data responsibly, and provides recourse when something goes wrong. This insight informs the design of PRV Nurse's verification pipeline and administrative oversight module.

2.3 Credential Verification in Professional Services

Credential verification is the process of confirming that a professional holds the qualifications they claim. In nursing, this includes confirming nursing qualification documents, institutional training records, and registration with a licensing body such as the Nursing and Midwifery Council (NMC) of Ghana [16]. Digital platforms face the challenge of conducting this verification without a direct relationship with licensing bodies, often relying on document upload and human review.

Korir [17] studied digital credential verification systems in East Africa and found that a hybrid approach, combining document uploads reviewed by human administrators with database checks against available institutional records, achieved higher accuracy than either method alone. A tiered badge system, where nurses are marked as 'document uploaded', 'verified', or 'background cleared', gives clients a clear signal of trust level and allows the platform to activate nurses in stages.

In the context of PRV Nurse, the verification pipeline follows this tiered model. Nurses submit qualification documents, license numbers, and identity verification. An administrator reviews and cross-references with available records. Background checks are

initiated separately. The nurse profile displays a verification badge that updates as each stage is cleared. This approach is consistent with best practices identified in the literature and practical for the Ghanaian context where real-time API integration with licensing bodies may not yet be feasible [17].

2.4 Machine Learning in Healthcare Recommendation

Recommendation systems have a well-established role in e-commerce and media platforms, but their application to healthcare provider matching is a growing and important area of research. In the context of home nursing, a recommendation system must balance multiple factors: clinical skills, location, availability, user ratings, cost, and personal preferences such as nurse gender or language.

Luo et al. [8] proposed a content-based filtering approach for nurse-patient matching in a home care context. Their model represented nurse profiles and patient care needs as feature vectors and used cosine similarity to rank suitable nurses. The approach performed well when profile data was rich but degraded for new nurses with limited history, a classic cold-start problem.

Seneviratne et al. [18] applied collaborative filtering to a healthcare provider matching problem, building a model that used historical booking and rating data to identify latent preference patterns. Their model improved match quality over time as more interaction data accumulated, addressing the weakness of purely content-based approaches. They also noted the ethical importance of ensuring that recommendation systems do not systematically disadvantage providers based on demographic characteristics.

For PRV Nurse, a hybrid recommendation approach is proposed, combining content-based filtering for cold-start cases (new nurses or new patients with no history) with collaborative filtering elements that improve personalization over time. A gradient-boosted tree model (specifically XGBoost) is selected as the primary ranking model because it handles mixed feature types well, is interpretable, and performs strongly on tabular data [19]. The features used include skill overlap score, distance penalty, availability match, rating average, booking completion rate, and interaction history signals.

2.5 Mobile Health (mHealth) in Ghana and Sub-Saharan Africa

Mobile health, or mHealth, refers to the use of mobile devices and wireless technologies to deliver health services and information. Ghana has seen significant mHealth activity since the mid-2010s, primarily in areas of disease surveillance, maternal health reminders, and community health worker support [20]. Ghana's mobile penetration rate exceeded 130% as of 2022 (accounting for multiple SIMs per person), and mobile money adoption, particularly through MTN MoMo, has reached a large portion of the adult population [21].

These conditions make Ghana a favorable environment for a mobile-first health platform. However, mHealth research in Africa also highlights important design constraints. Adepoju [7] notes that data costs, low-end device compatibility, and inconsistent network coverage in peri-urban areas mean that platforms must be designed to be lightweight, functional on mid-range devices, and tolerant of intermittent connectivity.

PRV Nurse addresses these constraints by building the frontend as a progressive web application (PWA) using React and Tailwind CSS, enabling offline-tolerant behavior, fast load times, and compatibility with a wide range of Android devices. The backend API is designed to be stateless and cacheable, reducing data usage for repeated requests.

2.6 Comparative Analysis of Related Systems

Table 2.1 presents a comparative analysis of existing home care and nurse placement platforms, situating PRV Nurse within the existing landscape. The comparison evaluates each system on the dimensions most relevant to the Ghanaian context.

Platform	Region	Credential Verify	ML Matching	Mobile Payment	Admin Portal	Open Source
Care.com	USA/Global	Partial	Yes	No (Credit Card)	Yes	No
Homage	Southeast Asia	Yes	Yes	Bank Transfer	Yes	No

Platform	Region	Credential Verify	ML Matching	Mobile Payment	Admin Portal	Open Source
TaskUs Health	Philippines	Yes	No	Bank/Card	Yes	No
Nurse Anywhere	Australia	Yes	No	Bank	Limited	No
PRV Nurse (Proposed)	Ghana	Yes (Tiered)	Yes (Hybrid ML)	Mobile Money	Full	Yes (Academic)

Table 2.1: Comparative analysis of home nursing platforms

As shown in Table 2.1, PRV Nurse is unique in its combination of tiered credential verification, hybrid machine learning matching, mobile money payment integration, and a full administrative portal designed for the Ghanaian context. Existing platforms that offer ML-based matching, such as Homage, operate in markets with more structured data infrastructure and banking systems. PRV Nurse adapts these capabilities to the Ghanaian mobile-money-first economy.

2.7 Research Gap

The literature review reveals a clear gap: while digital home care platforms exist globally and machine learning matching has been applied in healthcare contexts, there is no published system designed specifically to mobilize unposted qualified nurses in a sub-Saharan African context into a transparent, verified, and ML-powered home care marketplace. Existing platforms assume banking infrastructure, stable regulatory databases, and markets where professional licensure is easily verifiable digitally. PRV Nurse addresses this gap with a context-appropriate design that works within Ghana's current infrastructure while creating a foundation that scales as that infrastructure matures.

Furthermore, the combination of a machine learning recommendation engine with a structured nurse verification and booking lifecycle, served through a mobile-first

progressive web application and backed by mobile money, represents a technically novel integration not yet described in the academic literature for this context. This project contributes both a working design and an evaluation methodology that can inform future efforts in similar markets.

CHAPTER 3: METHODOLOGY

3.1 Research Design and Approach

This project uses a design science research (DSR) methodology, which is appropriate for engineering projects that create and evaluate IT artifacts [22]. DSR involves iterating through design, build, and evaluate cycles, using each iteration to refine the artifact. For PRV Nurse, three design iterations are planned. The first produces low-fidelity wireframes and system architecture models. The second produces a functional prototype with the core booking and matching flows. The third incorporates user testing feedback and refines the ML model with simulated data.

Data collection for the design phase involves structured interviews with five unposted nurses and five households with recent home care needs in Accra, conducted to validate assumptions about feature requirements, pricing expectations, and barriers to adoption. These findings directly informed the data models and feature priorities described in this chapter.

3.2 System Architecture Overview

PRV Nurse follows a three-tier client-server architecture with an additional machine learning service layer. Figure 3.1 illustrates the high-level architectural diagram. The presentation tier consists of the React and Tailwind CSS frontend served as a Progressive Web Application. The application tier consists of a Node.js REST API that handles business logic, authentication, and data access. The data tier consists of a MongoDB database. The ML service is a separate Python-based microservice that exposes a REST endpoint for recommendation queries.

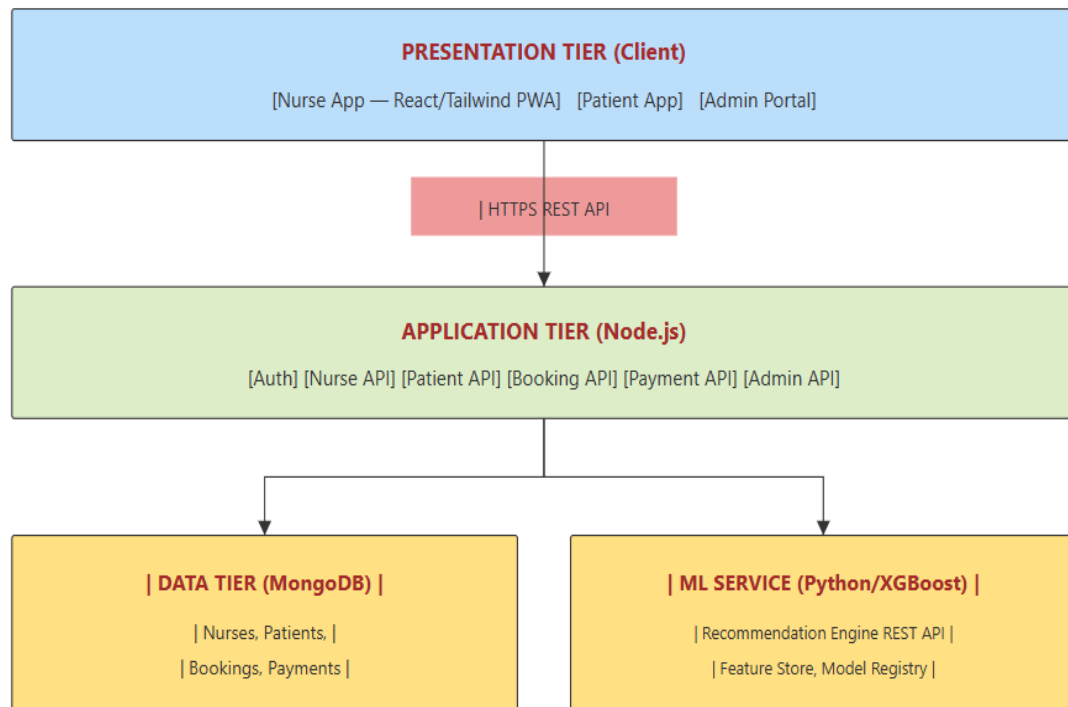


Figure 3.1: High-level system architecture of PRV Nurse

The architecture separates concerns cleanly: the frontend handles rendering and user interaction, the Node.js API handles authentication, authorization, business logic, and data orchestration, and the ML service handles computationally intensive recommendation queries independently. This separation allows each layer to be scaled, updated, or replaced without affecting the others.

3.3 Technology Stack

The technology decisions for PRV Nurse are driven by three criteria: developer productivity (leveraging widely available skills in Ghana's developer community), mobile-first performance, and open-source availability. Table 3.1 summarizes the full technology stack.

Layer	Technology	Justification
UI Design	Figma	Industry-standard prototyping with shared design tokens and component library
Frontend	React 18 + Tailwind CSS v3	Component-based UI with utility-first styling; PWA support; large ecosystem
State Management	Redux Toolkit + React Query	Predictable state for auth/session; server state caching for API data
Backend	Node.js (Express.js)	Non-blocking I/O ideal for real-time booking events; JSON native; fast dev cycle
Database	MongoDB Atlas	Flexible schema for heterogeneous nurse/patient profiles; geospatial indexing built-in
Authentication	JWT + bcrypt	Stateless auth suitable for mobile clients; role-based access control
ML Service	Python + XGBoost + FastAPI	High-performance gradient boosting for recommendation ranking; REST-exposed via FastAPI
File Storage	AWS S3 / Cloudinary	Secure credential document storage with access control; CDN delivery for profile photos
Payments	MTN MoMo API	Ghana's dominant mobile money provider; REST API for collection and disbursement
Maps / Location	Google Maps API	Geocoding, distance matrix for nurse-patient proximity matching
Notifications	Firebase Cloud Messaging	Push notifications for booking updates, emergencies, and reminders

Table 3.1: PRV Nurse technology stack

3.3.1 Figma for UI Design

Figma was used as the primary design and prototyping tool. A shared component library was created with consistent design tokens for colours (navy primary, red accent), typography (Inter font family), spacing scale, and button variants. Screens were designed for mobile-first at 375px width and adapted for desktop at 1440px. The Figma prototypes served as the specification for frontend development, reducing ambiguity and enabling asynchronous review of design decisions before code was written [23].

3.3.2 React and Tailwind CSS Frontend

React was selected as the frontend framework for its component reusability, large developer community, and strong support for progressive web application features. Tailwind CSS provides a utility-first styling approach that integrates cleanly with React's JSX syntax, avoiding the overhead of maintaining a separate CSS file hierarchy. The combination enables rapid UI development while maintaining visual consistency [24].

The frontend application is structured into three distinct portals sharing a common component library: the Nurse Portal, the Patient Portal, and the Admin Portal. Each portal is accessed by a separate authenticated route and role check. React Router v6 handles client-side navigation. React Query manages server state, handles API caching, and provides loading and error states for all data-fetching operations.

3.3.3 Node.js Backend

The backend is implemented as a RESTful API using Node.js and Express.js. Express was chosen for its minimal footprint, extensive middleware ecosystem, and widespread familiarity. The API is organized into resource-based routers: /auth, /nurses, /patients, /bookings, /payments, /admin, and /recommendations. Each router delegates to a controller layer, which calls service functions that contain business logic. This three-layer structure (route → controller → service) keeps responsibilities clear and makes unit testing straightforward [25].

Mongoose is used as the MongoDB ODM (Object Document Mapper), providing schema validation at the application layer, virtuals, and middleware hooks. JWT tokens are issued

on login with a 24-hour expiry. Role-based access control middleware is applied to all protected routes, enforcing that nurses cannot access patient records, patients cannot access nurse admin controls, and only administrators can approve verifications.

3.4 System Modules and Components

PRV Nurse is composed of six primary modules, each serving a distinct function in the overall platform. Figure 3.2 shows the module interaction diagram.

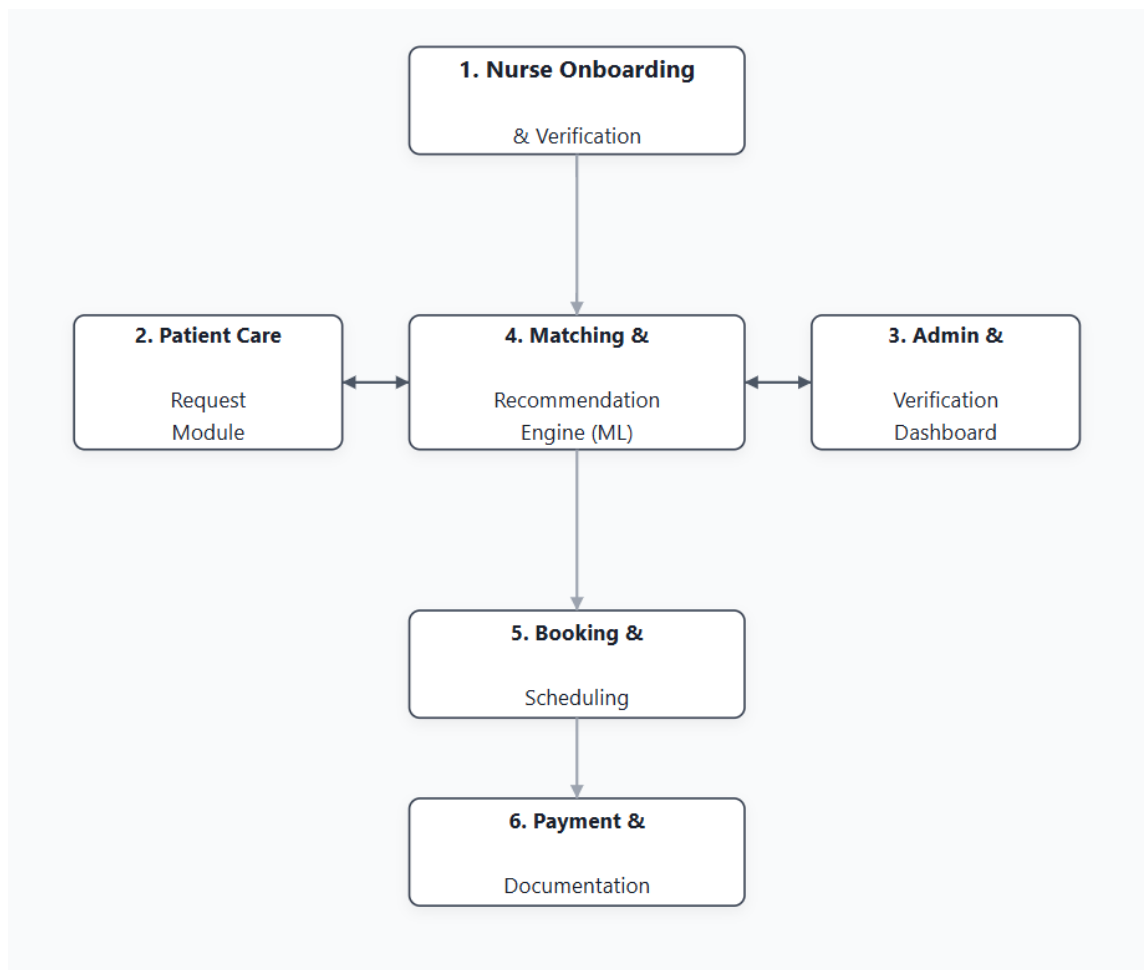


Figure 3.2: Module interaction diagram for PRV Nurse

3.4.1 Nurse Onboarding and Verification Module

The nurse onboarding module handles all steps from initial registration to activation on the platform. It is divided into five sequential stages: basic identity collection, identity verification, professional qualifications, work experience and specializations, and availability and pricing setup. A nurse cannot receive bookings until the administrator marks their profile as verified.

Table 3.2 shows the nurse profile data schema captured during onboarding, derived from the platform's nurse requirements specification.

Category	Field	Description / Purpose
Basic Identity	Full Name, Gender, DOB	Core identity for display and age-based matching
	Phone, Email, Photo	Contact details; photo mandatory for trust signal
ID Verification	National ID / Passport	Document upload for identity confirmation
	Passport Photo, Address	Photo must match profile photo; address verification
Qualifications	Degree, Institution, Year	Nursing qualification level and source institution
	License Number, Expiry	NMC registration details; used for renewal reminders
	Certifications	RN, Midwifery, ICU, Pediatrics, etc.
Experience	Years, Hospitals, Roles	Work history for matching and display
	Specializations	Elderly care, post-surgery, pediatric, emergency
Services & Skills	Service List	Medication admin, wound dressing, vitals, home visits

Category	Field	Description / Purpose
Availability	Days, Time Slots	Morning/Afternoon/Night; emergency availability flag
Pricing	Hourly/Daily/Weekly/Emergency Rate	Rate card visible to patients during matching
Trust Signals	Verification Badge, Background Check	Tiered badge updated by admin after each check stage

Table 3.2: Nurse profile schema and field descriptions

The verification flow follows the sequence shown in Figure 3.3. After document submission, an administrator reviews uploads within 48 hours. If a license number is provided, the administrator manually cross-references with available NMC records. Background check initiation is a separate admin action. The nurse's badge progresses from 'Pending' to 'Documents Verified' to 'Background Cleared' as each stage completes.

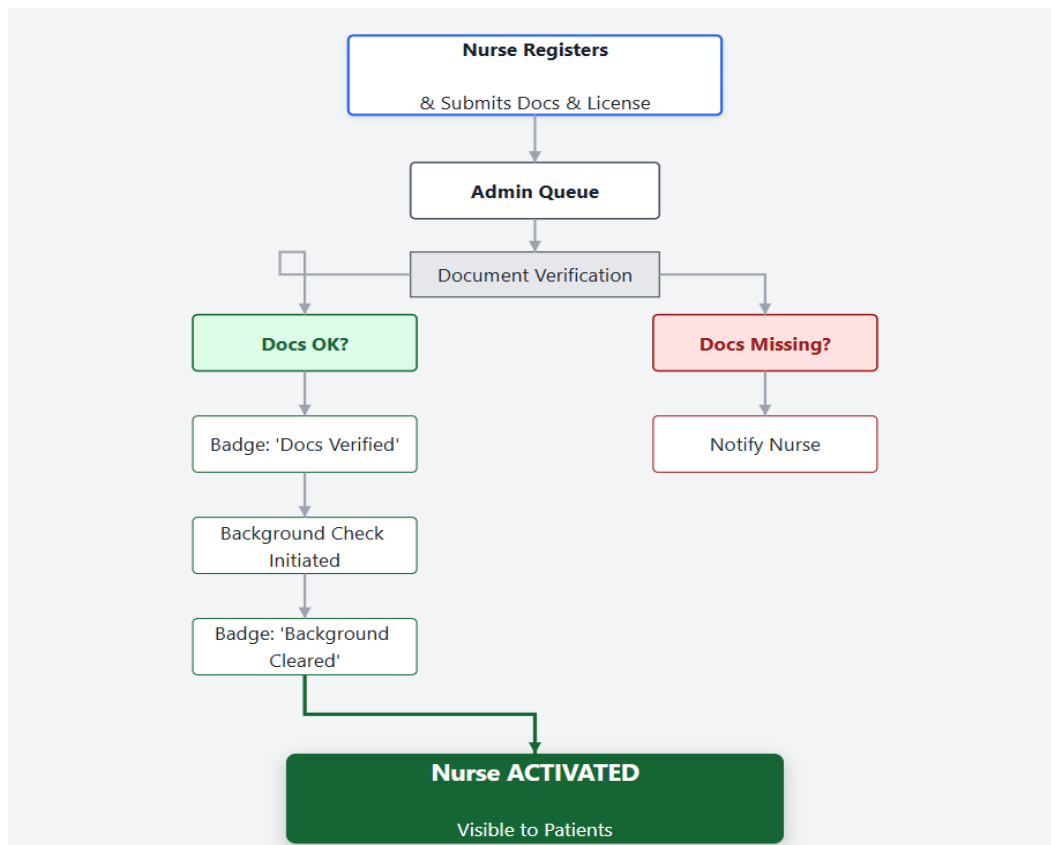


Figure 3.3: Nurse verification workflow

3.4.2 Patient Request and Care Module

The patient module allows individuals or their designated caregivers to create a care request. Table 3.3 shows the patient profile and request schema.

Category	Field	Notes
Identity	Full Name, Gender, DOB	Age used for matching (elderly patients may need specialised nurses)
Location	GPS, Address, Landmark	Used for proximity matching; landmark helps nurse navigate
Medical Info	Primary Condition	e.g. elderly care, post-surgery, chronic illness
	Medications, Allergies	Shared with matched nurse prior to first visit
	Mobility Status	Bedridden, limited mobility, or independent — affects task scope
Care Requirements	Type of Care	Home care, elderly, post-surgical, emergency support
	Specific Tasks	Medication admin, wound care, vitals, physiotherapy assist
	Preferred Nurse Gender	Optional preference; used as soft constraint in matching
Schedule	Start Date, Duration, Time Slots	Hourly/daily/weekly/long-term; morning/afternoon/night
Emergency Contact	Name, Relationship, Phone	Mandatory; notified in case of incident during care

Category	Field	Notes
Payment	Mobile Money Number	Wallet number for payment initiation

Table 3.3: Patient profile and care request schema

3.4.3 Admin Dashboard

The administrative dashboard is a React-based portal accessible only to platform administrators. It provides a central view of platform operations, drawing on the dashboard design shown in the interface prototype. Key panels include an overview of active nurses, total patients, bookings today, revenue today, and emergency requests. A booking trends chart shows the past 14 days of booking and completion data. A nurse availability heatmap shows supply coverage across days and time slots.

The pending verifications panel is a critical operational tool. It lists nurses awaiting review with their document status (uploaded, missing, pending) and background check status (pending, cleared). Administrators can approve or reject each nurse directly from the dashboard. A patient flags tab surfaces any patient accounts that have triggered safety or complaint workflows.

The platform health panel monitors API uptime, payment success rate, failed payments count, average nurse-patient match time, and open disputes. These metrics enable administrators to identify and respond to platform degradation quickly.

3.4.4 Matching and Recommendation Engine

The matching and recommendation engine is the technical centrepiece of PRV Nurse. It is responsible for ranking available and verified nurses against a patient care request and returning a prioritized shortlist. The engine runs as a Python microservice, called by the Node.js backend whenever a patient submits a care request or browses available nurses.

The matching process operates in two stages: candidate generation and candidate ranking. In the candidate generation stage, the database is filtered to return only nurses who meet

hard constraints: they must be verified, available on the requested dates and times, willing to travel to the patient's location, and possess at least one of the required care specializations. This stage uses MongoDB geospatial queries for proximity filtering and indexed availability checks.

In the candidate ranking stage, the filtered nurses are scored by the XGBoost model and returned in ranked order. Table 3.4 shows the features used in the ranking model.

Feature	Type	Description
Skill Overlap Score	Numeric [0-1]	Jaccard similarity between nurse skills and patient required tasks
Distance Penalty	Numeric [km]	Haversine distance from nurse home to patient location; normalized
Availability Match	Binary / Numeric	Exact match on day and time slot; partial match also scored
Average Rating	Numeric [1-5]	Mean rating from past patient reviews
Completion Rate	Numeric [0-1]	Completed bookings / total accepted bookings
Experience (Years)	Numeric	Total years of nursing experience; log-transformed
Specialization Match	Binary	Nurse has explicit certification matching patient's condition
Gender Preference Match	Binary	Nurse gender matches patient's stated preference (if any)
Price in Budget	Binary	Nurse rate within patient's indicated budget range
Historical Interaction	Numeric [0-1]	Collaborative signal: other patients like this one rated this nurse highly

Table 3.4: Features used in the XGBoost recommendation model

The model is trained using historical booking and rating data collected during the pilot phase. For cold-start situations, where a nurse is new and has no rating or completion history, the historical interaction feature is set to a neutral prior (0.5) and the model relies primarily on content-based features. As the nurse accumulates interaction data, the collaborative signal gains weight.

Figure 3.4 illustrates the data flow through the recommendation engine from patient request to ranked nurse list.

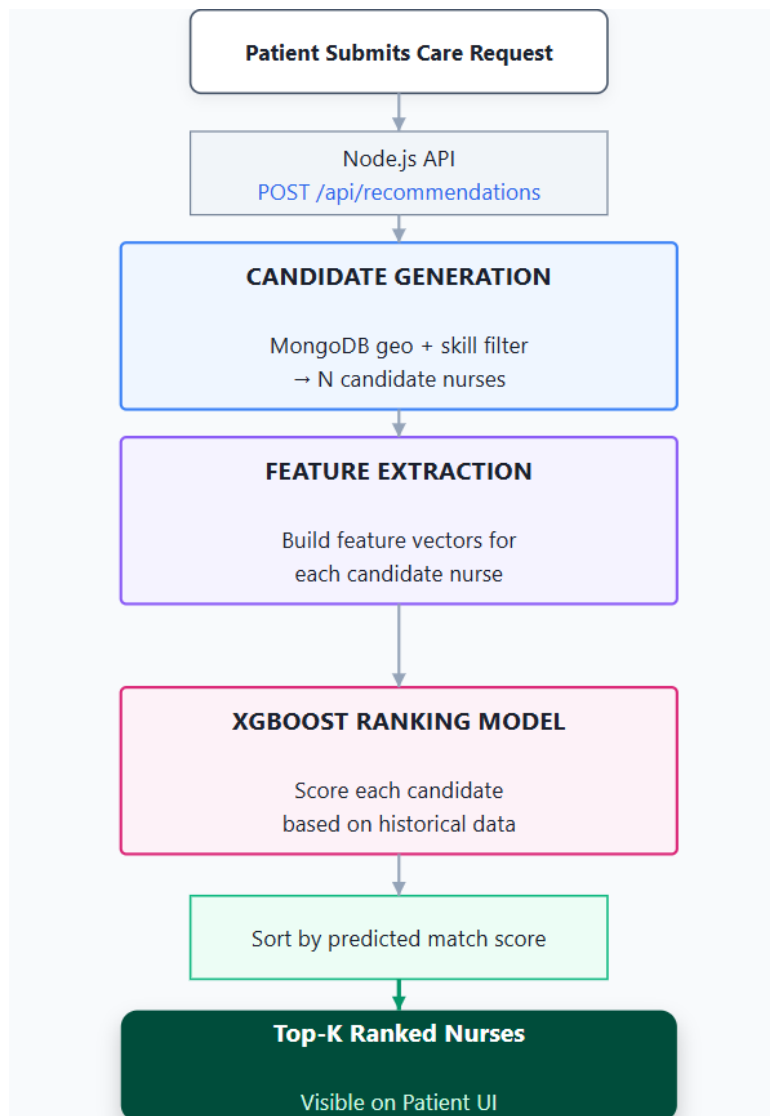


Figure 3.4: Recommendation engine data flow

3.4.5 Booking and Scheduling Module

Once a patient selects a nurse from the recommendation results, the booking module handles the request lifecycle. The booking state machine has six states: Requested, Accepted, In Progress, Completed, Cancelled, and Disputed. Figure 3.5 illustrates these state transitions.

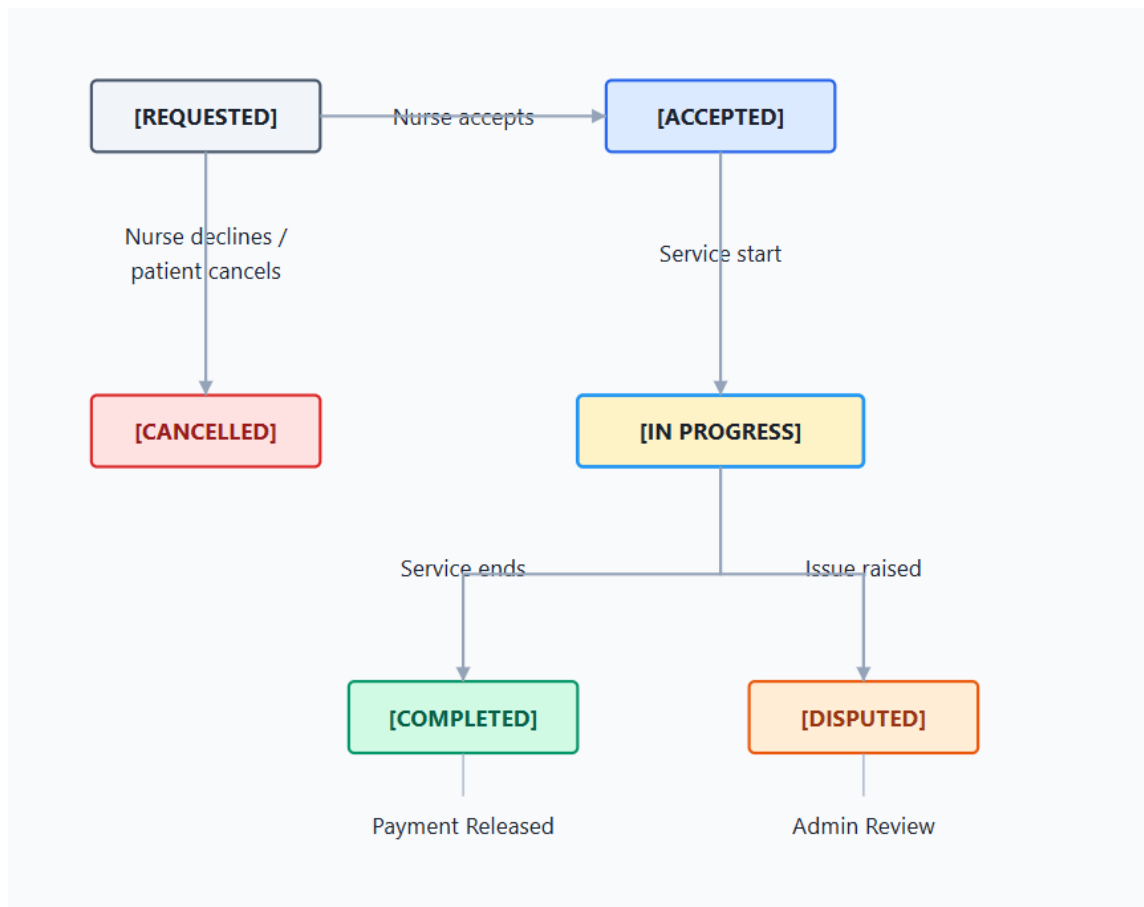


Figure 3.5: Booking state machine

When a booking moves to 'Accepted', a calendar entry is created for the nurse and a confirmation notification is sent to both parties via Firebase Cloud Messaging. For recurring bookings (daily or weekly), the system creates a series of linked booking records so that any individual session can be cancelled without affecting the series.

3.4.6 Payment and Documentation Module

Payments are processed using the MTN Mobile Money API. When a booking is marked 'Completed', the system initiates a payment request from the patient's registered mobile money wallet to the platform's collection wallet. After a 48-hour hold period (to allow disputes), the funds are disbursed to the nurse's registered mobile money number minus a platform service fee.

The care documentation sub-module allows nurses to record a visit note at the end of each service session. Fields include vitals recorded (blood pressure, temperature, pulse rate, SPO2), medications administered, observations, and any concerns to flag to the patient's emergency contact or next scheduled nurse. These records are stored per patient and accessible to the patient and their emergency contact, creating a basic care continuity record.

3.5 Data Flow Architecture

Figure 3.6 shows the end-to-end data flow from nurse registration to completed booking, illustrating how data moves between the frontend, Node.js API, MongoDB, and the ML service.

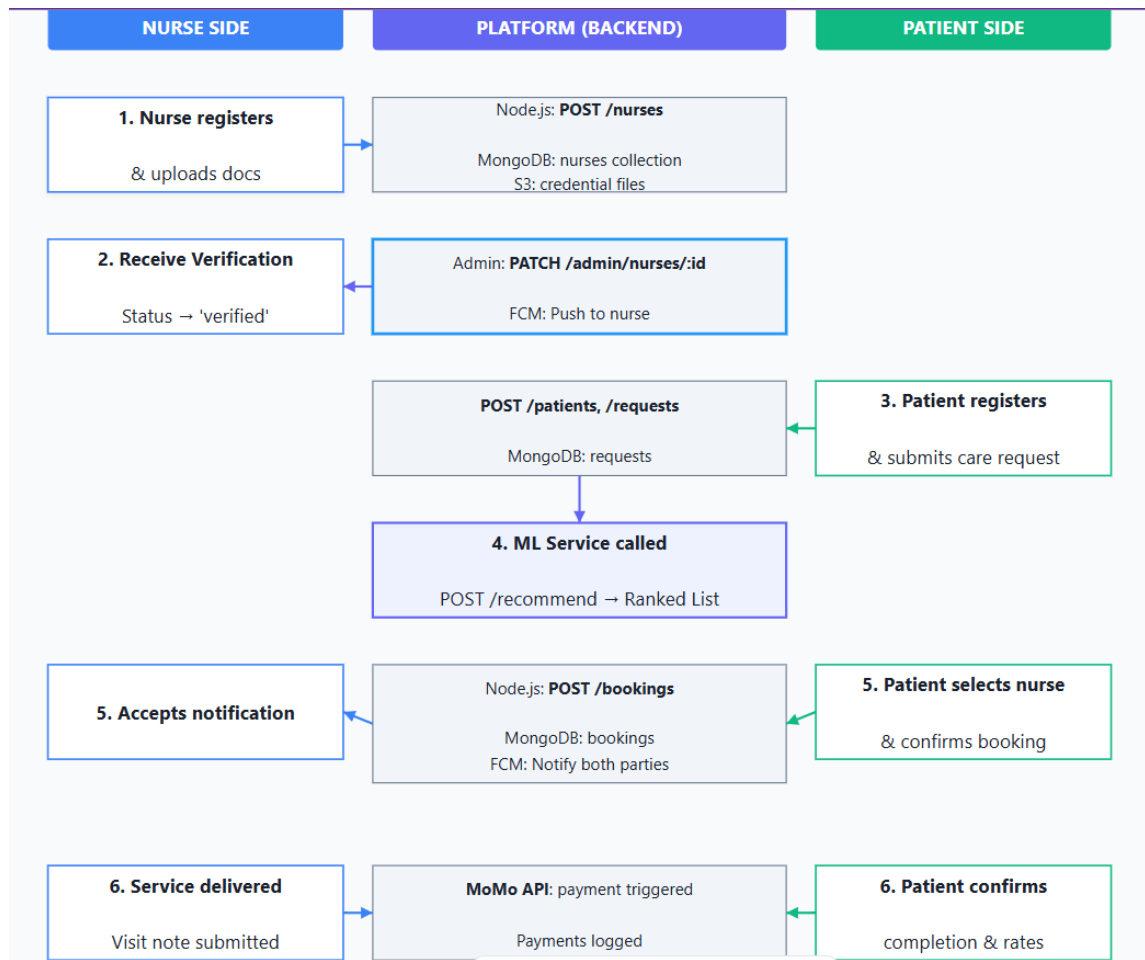


Figure 3.6: End-to-end data flow diagram

3.6 Database Design

PRV Nurse uses MongoDB, a document-oriented NoSQL database, as its primary data store. MongoDB was chosen for its flexible schema, which accommodates the heterogeneous structure of nurse and patient profiles without requiring rigid table schemas. Its built-in geospatial indexing (2dsphere indexes) directly supports the location-based filtering needed in the candidate generation stage. Table 3.5 describes the primary collections.

Collection	Key Fields	Purpose
nurses	_id, status, skills[], location{}, availability{}, ratings	Nurse profiles; 2dsphere index on location for geo queries
patients	_id, careNeeds[], location{}, medicalInfo{}	Patient profiles and care request history
bookings	_id, nurseId, patientId, status, schedule{}, visitNotes[]	Booking lifecycle records; status enum for state machine
payments	_id, bookingId, amount, status, momoRef, timestamp	Payment records linked to bookings; MoMo transaction reference
reviews	_id, nurseId, patientId, bookingId, rating, comment	Post-booking reviews; aggregated into nurse average rating
verifications	_id, nurseId, docUrls[], status, adminNotes	Tracks document upload and verification status per nurse
notifications	_id, userId, type, message, read, createdAt	In-app and push notification records

Table 3.5: MongoDB collection schema summary

Relationships between collections are maintained through references (nurseId, patientId, bookingId) rather than embedding, because the primary access patterns are by individual entity (look up a specific nurse, a specific booking) rather than by aggregation. Embedding would create overly large documents for active nurses with many bookings.

3.7 Security and Privacy Design

Handling personal and medical data places significant responsibility on the platform. Several security and privacy controls are incorporated into the design.

Authentication uses JSON Web Tokens (JWT) issued by the Node.js API after successful login. Tokens carry a role claim (nurse, patient, admin) used by middleware to enforce role-based access control on all API routes. Tokens expire after 24 hours, and a refresh token mechanism enables silent re-authentication. Passwords are hashed using bcrypt with a work factor of 12 before storage.

Medical data (medications, medical history, allergy records) stored in the patient collection is encrypted at rest using MongoDB's field-level encryption. Only the matched nurse assigned to a booking, and the patient themselves, can decrypt and view this data. Administrators can access anonymized summaries for oversight purposes but cannot view individual patient medical records.

Credential documents uploaded by nurses (certificates, license scans, ID copies) are stored in AWS S3 with signed URL access. Documents are never publicly accessible; all retrieval requests pass through the Node.js API, which checks the requesting user's role and booking relationship before generating a time-limited signed URL.

Audit logs are maintained for all administrative actions, including verification approvals and rejections, dispute resolutions, and any access to patient medical data. This log supports accountability and provides evidence in case of disputes or regulatory inquiries [15].

3.8 System User Flows

This section describes the primary user flows for each role on the platform.

3.8.1 Nurse Registration and Activation Flow

A new nurse downloads the PRV Nurse app (PWA) and selects 'Register as Nurse'. The multi-step registration form collects basic identity information on screen one, professional qualifications and license number on screen two, work experience and specializations on screen three, and availability and pricing on screen four. On the final screen, the nurse

uploads credential documents and submits. The nurse receives an email and push notification confirming receipt and is shown a badge reading 'Verification Pending'. Within 48 hours, the admin reviews and either approves (badge updates to 'Verified') or sends back a request for missing documents. Once fully verified, the nurse is activated and their profile becomes visible to patients.

3.8.2 Patient Booking Flow

A patient or caregiver registers on the patient portal. They create a care request by specifying the type of care, required tasks, medical notes, schedule, and location. On submission, the recommendation engine returns a ranked list of up to ten suitable nurses, each shown with their name, photo, verification badge, specializations, ratings, and rate. The patient reviews profiles, optionally views full profiles for detailed information, and selects their preferred nurse. The booking request is sent to the nurse, who has 30 minutes to accept or decline. On acceptance, both parties receive confirmation notifications and calendar entries.

3.8.3 Emergency Request Flow

Emergency requests bypass the standard booking flow. A patient or caregiver triggers an emergency request from the main patient dashboard. The system immediately queries for the nearest verified nurse with emergency availability enabled and sends a push notification and SMS. The nurse can accept within five minutes; if not accepted, the next nearest nurse is notified. The admin dashboard highlights all active emergency requests in a dedicated panel for monitoring. If no nurse accepts within 15 minutes, the system escalates with guidance to contact emergency services.

3.9 Evaluation Plan

The evaluation of PRV Nurse will be conducted across two phases. Phase one is a prototype evaluation using usability testing with representative users: five unposted nurses and five patients or caregivers recruited from community health networks in Accra. Participants will complete a series of defined tasks (nurse registration, submitting a care

request, reviewing a recommendation) while thinking aloud. Task completion rates, error rates, and System Usability Scale (SUS) scores will be recorded.

Phase two is a pilot deployment with a small cohort of 20 verified nurses and an open patient population for a period of eight weeks. Table 3.6 shows the evaluation metrics that will be tracked during the pilot.

Category	Metric	Target / Benchmark
Employment	Nurses onboarded and active	20 nurses; ≥ 15 active within 4 weeks
	Average earnings per nurse	Monitored; no pre-set target for pilot
Access	Care requests filled	$\geq 80\%$ of submitted requests matched and accepted
	Time to match	Average ≤ 5 minutes from request to nurse notification
Quality	Patient satisfaction (NPS)	NPS ≥ 50 at end of pilot
	Booking completion rate	$\geq 90\%$ of accepted bookings completed
ML Performance	Recommendation relevance (CTR)	Click-through rate on top-3 recommendations $\geq 60\%$
	Model A/B test	ML ranking vs. random ordering; CTR improvement $\geq 15\%$
Safety	Incident reports	Zero critical safety incidents; any incidents documented
Platform Health	API uptime	$\geq 99.5\%$
	Payment success rate	$\geq 95\%$ of initiated payments completed

Table 3.6: Pilot evaluation metrics and targets

Statistical analysis of pilot data will use paired t-tests to compare recommendation click-through rates between the ML-ranked condition and a random ordering control condition. Patient satisfaction and booking completion rate will be compared against benchmarks reported for comparable platforms in developing market contexts [14]. All data collection will comply with relevant data protection principles, with written informed consent from all participants.

3.10 Risk Assessment and Mitigation

Table 3.7 summarizes the key risks identified for the PRV Nurse project and the mitigation strategies assigned to each.

Risk	Likelihood	Mitigation
Safety incident during home care visit	Low-Medium	Strict verification; emergency escalation protocol; liability waiver; incident reporting
Low nurse adoption / cold-start problem	Medium	Partner with nursing schools and associations; incentive program for first 50 nurses
Mobile Money payment failures	Medium	Retry logic; fallback to manual disbursement; clear dispute channel
Credential fraud (fake documents)	Low-Medium	Multi-stage verification; admin review; spot checks with NMC; badge revocation mechanism
Data privacy breach	Low	Encryption at rest and in transit; field-level encryption for medical data; audit logs
Poor network connectivity affecting app usability	Medium	Progressive Web App with offline-tolerant design; minimal payload API responses

Risk	Likelihood	Mitigation
ML model bias (unfair to new nurses)	Medium	Cold-start prior; fairness constraint in ranking (minimum exposure for new nurses)

Table 3.7: Risk assessment and mitigation strategies

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